

Pappu Jha

Hattiesburg, Mississippi, United States | jhapappu7165@gmail.com | pappu.jha@usm.edu

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Education

The University of Southern Mississippi (USM)	Hattiesburg, Mississippi
Bachelor of Science (BS) in Computer Science	GPA: 4.0/4.0
Minors: Economic Data Analysis, Information Security	Graduation: May 2027
Honors Keystone Scholar, TEDx Southern Miss 2025 Speaker	
Coursework: <u>Designing Solutions for Defense</u> , Data Structures & Algorithms, Econometrics, Machine Learning in Cybersecurity, Web Application Security, Cybersecurity Principles, Probability & Statistics	

Work Experiences

IT Data Analyst Intern , Jones Capital – Hattiesburg, MS	Sept 2025 – Present
- Designed and modified 10+ automation workflows in Jira Service Management (JSM) across over 8 portfolio companies, reducing service request resolution time by 25% and aligning delivery with SLA targets. - Built 30+ dynamic gadgets in JSM and Excel using JQL filters to monitor real-time performance of 5 IT agents, resulting in a 30% increase in team efficiency and improved KPI tracking. - Developed 5 automation pipelines in n8n, created 20+ pages of process documentation in Confluence, and published 5 Scribe tutorials, reducing manual reporting time by 40%. - Resolved 10+ field-level discrepancies in more than 2000 JSM work items of all time by troubleshooting, leading to a 100% reduction in recurring ticket errors in the following weeks.	
Data Analytics & Business Intelligence Intern (Aviation Strategy) , Port Authority of New York and New Jersey – New York, NY	May 2026 – August 2026
- Engineered and deployed machine learning regression, time-series, and deep learning models using Python's PySpark on Databricks to estimate hourly passenger volumes at Newark Airport (EWR), integrating more than 2 third-party vendor datasets and achieving an R^2 above 80% — closing a critical gap in PANYNJ's access to real-time hourly passenger forecasting. - Designed Power BI dashboards translating hourly demand forecasts into actionable operational insights, enabling data-driven resource allocation at one of the nation's busiest airports (47M+ passengers in 2025). - Developed and deployed an internal Legislation Tracker that monitors and filters up to 250 federal, New York, and New Jersey legislative items, with department-specific categorization, update tracking, and analyst notes to streamline policy monitoring for Port Authority stakeholders.	
Data Engineer/Research Assistant , Institute for Advanced Analytics and Society – Hattiesburg, MS	May 2025 – May 2026
- Built AQUAVIEW, a Python library exposing more than 7 API endpoints for streamlined access to oceanographic data for NOAA (National Oceanic and Atmospheric Administration), downloaded over 1000 times. - Processed 17M+ data from 125 years of the World Ocean Database, generated metadata in MongoDB, indexed in the cloud, and integrated into the UI via Elasticsearch, contributing to NOAA's AQUAVIEW project. - Enhanced the user interface of World Ocean Database to handle edge cases, including invalid date ranges and out-of-bound latitude/longitude values, improving overall user experience. - Developed an LLM-powered classification tool to assign Platform IDs for Uncrewed Marine Systems using metadata in real-time, streamlining NOAA's platform designation across 20M+ data points in 10+ sources. - Created an adapter for the National Data Buoy Center (NDBC) historical and real-time data from inactive and active stations, enabling easy access to over 30 years of NDBC datasets.	

- Web Developer**, Roy Howard Community Journalism Center – Hattiesburg, MS **Oct 2024 – May 2025**
- Developed and maintained the organization’s content-heavy website using **WordPress (SNO CMS)**, improving user engagement by **35%** through targeted UI/UX redesign and responsive layouts.
 - Enhanced visibility by optimizing **SEO metadata**, improving page load speed by **40%**, and indexing high-traffic pages on **Google Search Console**.

Research Experience

- Undergraduate Research Assistant**, Cyber Innovation Lab – Hattiesburg, MS **Sept 2024 – Present**
Dr. Nick Rahimi, Associate Professor, USM School of Computing Sciences & Computer Engineering
- First-authored **Adversarial Machine Learning for Robust Password Strength Detection**, developing 5 **machine learning** models to classify adversarial samples across three categories, improving accuracy by **20%**.
 - Co-authored **Social Engineering Attack Detection in Corporate Communications**, evaluating **machine learning** (Logistic Regression, Naive Bayes) and **deep learning** (BERT) models for phishing email classification and achieving 90%+ accuracy on real-world datasets.
 - Led and mentored an **8-student interdisciplinary research team** in the **Computing Meets Green Chemistry** program, guiding development of data visualization and machine learning tools for nanomaterials synthesis data while coordinating cross-disciplinary research with Chemistry students and faculty.

Teaching Experiences

- Learning Assistant**, Center for Faculty Development – Hattiesburg, MS **Aug 2024 – May 2025**
Dr. Timothy Gutmann, Assistant Professor of Religion, USM School of Humanities
- Served as Learning Assistant for **REL 131 (Comparative Religion)**, co-developing the course plan with the professor and leading implementation across 40+ students per semester over 2 semesters.
 - Graded student performance in team-based work, an integral component of the course, coordinating 50+ group activities and providing structured feedback that shaped instructional adjustments throughout the term.
 - Served as the primary liaison between students and the professor, surfacing recurring student concerns and relaying performance insights that informed real-time course improvements.

- Academic Tutor**, Trinity International SS & College – Kathmandu, Nepal **July 2022 – July 2023**
Cambridge International AS & A-Level (University of Cambridge), Physical Sciences
- Tutored **Cambridge AS & A2-Level students** in Computer Science, Physics, Chemistry, and Mathematics across 4 classes (30 students per class), reaching up to 120 students during a gap-year teaching engagement.
 - Re-taught and clarified core concepts across all four subjects tailored to individual student needs, designing focused review sessions to strengthen exam readiness and conceptual understanding.
 - Advised students on college admissions and career pathways alongside academic instruction, extending the tutoring role beyond subject content into long-term academic planning.

Creative Projects

- Boston Daddy** [Link](#)
Hack Harvard 2025, Harvard University **Oct 2025**
- Developed a **full-stack** smart city infrastructure platform that integrated energy, traffic, and weather data — achieving predictive analytics and actionable insights for 15+ buildings and 10+ intersections in real-time.
- Delagent** [Link](#)
MHacks 2025, University of Michigan **Sept 2025**
- Built an AI-based scheduling platform using **Fetch.AI uAgents framework**, **FastAPI**, and **SQLite**, resulting in automated 3-agent calendar coordination and negotiation with 80% reduction in manual scheduling overhead.

Regulating College Dormitories Digitally Using C++

Advisor: Mr. Rick Loggins, USM Faculty Member

[Link](#)
Nov 2023

- Built a dormitory management back-end system using all concepts from the introductory C++ course, streamlining student check-ins/outs and visitor logs, reducing manual tracking effort by 60%.
- Integrated rule enforcement features to automatically flag dorm violations, improving housing compliance oversight of 500+ students.

Publication & Presentation

Conference Proceeding

- **Jha, P.**, Hamid, H., Olukola, O., Dahal, A., & Rahimi, N. (2026). *Adversarial Machine Learning for Robust Password Strength Estimation*. In: Rahimi, N., Margapuri, V., & Golilarz, N.A. (Eds.) *Software and Data Engineering. SEDE 2025*. Communications in Computer and Information Science, vol. 2720. Springer, Cham. https://doi.org/10.1007/978-3-032-08649-5_18
- Yamali, S., **Jha, P.**, Kandel, B., Poudel, A., & Hill, J. (2026). World Ocean Database: Building a search engine for 200 years of ocean data. Oral presentation and poster presented at the UG2 (Underwater Glider User Group) Workshop 2026, St. Petersburg, FL.

Presentations

- USM Undergraduate Symposium on Research & Creative Activity – **First Place** (Computational Approaches category), **Adversarial Machine Learning on Robust Password Strength Estimation** (Apr 2025)
- Golden Idea Pitch Competition, **Top 8 Finalist** – USM Center for Entrepreneurship (May 2026)
- Ignition Pitch Competition – Mississippi Polymer Institute (May 2025 & April 2026)
- USM Undergraduate Symposium on Research & Creative Activity, **Social Engineering Attack Detection in Corporate Communications** (May 2026)

Skills

Programming: Python, C++, OOP, API, SQL, Web Development (HTML, CSS, JavaScript, WordPress)

Information Technology: Atlassian (Jira Service Management, Confluence), n8n, Power BI, Scribe, Microsoft 365 tools, Cloud Services, GitHub, Database (MongoDB), SharePoint, OneDrive, Jupyter Notebook, LLM/GenAI

Data Analytics: Databricks, Power BI, Python (PySpark, NumPy, Pandas, Scikit-learn, Seaborn), SQL, Machine Learning (Random Forest, Logistic Regression, Naive Bayes), Object Oriented Programming (OOP)

Leadership Experiences

International Ambassador/Mentor, USM Center for International Education

Dec 2023 – Present

- Supported 100+ international students by managing pre-arrival communication on **Unibuddy** and on-site onboarding to ensure a smooth transition to the United States.

Creative Director, The Entrepreneurship Society (E-Society)

Aug 2025 – May 2026

- Conducted marketing of the organization via social media, mass emailing, and distributing flyers, increasing the membership to more than 80 in a semester.

Director of Finance, USM Residence Hall Association

Apr 2024 – May 2025

- Formulated budgets for 8+ RHA events with 1000+ attendees, coordinating with board members and advisors to ensure successful execution and efficient resource allocation.

Honors & Awards

USM President's Lists: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026	Aug 2023 – May 2026
First Place, Undergraduate Symposium for Research and Creative Activity (UGS)	Apr 2025
International Ambassador of the Year, USM Center for International Education	Apr 2025
Innovation & Excellence Award, Roy Howard Community Journalism Center, USM	Apr 2025
Honors Keystone Scholarship, USM Honors College	Aug 2025 - May 2027
Academic Excellence Scholarship, The University of Southern Mississippi	Aug 2023 - May 2027
Top 500 Global Finalist, Rise For The World	Jan 2022